PHYC10004 SWOTVAC Material
Amendments to LMS-posted exams and/or solutions – 2008 – 2017
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2017 Exam

Q4(b): Some versions of this paper ask you to show that the value of dielectric constant is ~**45**. This value is in error, caused by multiplying the provided area by itself (assuming it was a length), and it should instead be **~225.**

2016 Exam

2015 Exam

Q2(b): The identity used in the solution is incorrect (and differs from that given in the exam). While the final result is correct, the steps taken are not.
Between steps 20 and 21, the substitution is made that sinθ = R/r. This is correct, but its application is not. The steps following should look like those describe in text HRW (10th Edition), section 29-1, p839:

$$\int dB = \frac{i\mu_0}{4\pi}\int \frac{Rds}{r^3}$$

Note that $r = \sqrt{s^2 + R^2}$ so

$$= \frac{i\mu_0}{4\pi}\int \frac{Rds}{(s^2 + R^2)^{3/2}}$$

Now using the provided integral (#19 in HRW, same as on exam booklet), and evaluate for *s* between 0 and infinity (by symmetry, half of the magnetic magnitude *B*, so define LHS as *B*/2):

$$\frac{B}{2} = \frac{i\mu_0 R}{4\pi}\int_0^\infty \frac{ds}{(s^2 + R^2)^{\frac{3}{2}}}$$

$$= \frac{\mu_o iR}{4\pi}\left[\frac{s}{R^2(s^2 + R^2)^{\frac{1}{2}}}\right]_0^\infty$$

$$= \frac{\mu_o i}{4\pi R}\left[\frac{s}{s\left(1 + \left(\frac{R}{s}\right)^2\right)^{\frac{1}{2}}}\right]_0^\infty$$

$$= \frac{\mu_o i}{4\pi R}\left[\frac{1}{\left(1 + \left(\frac{R}{s}\right)^2\right)^{\frac{1}{2}}}\right]_{s=0}^{s=\infty}$$

This integrand can be resolved as → 1, so:

$$\frac{B}{2} = \frac{\mu_0 i}{4\pi R} \rightarrow B = \frac{\mu_0 i}{2\pi R}$$

for the whole wire.

2011 Exam